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0580/23

May/June 2018

1 hour 30 minutes

Additional Materials: Electronic calculator Geometrical instruments
Tracing paper (optional)

READ THESE INSTRUCTIONS FIRST

DO NOT WRITE IN ANY BARCODES.

For π , use either your calculator value or 3.142.

The total of the marks for this paper is 70.

This document consists of **11** printed pages and **1** blank page.

- 1** One day in Chamonix the temperature at noon was 6°C .
At midnight the temperature was 11°C lower.

Write down the temperature at midnight.

..... $^{\circ}\text{C}$ [1]

- 2** Factorise.
 $w + w^3$

..... [1]

- 3** Liz takes 65 seconds to run 400 m.

Calculate her average speed.

..... m/s [1]

- 4** Complete the list of factors of 36.

1, 2,, 36 [2]

- 5** Increase \$22 by 15%.

\$..... [2]

- 6 (a)** Write 209 802 correct to the nearest thousand.

..... [1]

(b) Write 4123 correct to 3 significant figures.

..... [1]

- 7 The probability that Kim wins a game is 0.72 .
In one year Kim will play 225 games.

Work out an estimate of the number of games Kim will win.

..... [2]

- 8 (a) Write 4.82×10^{-3} as an ordinary number.

..... [1]

(b) Write 52 million in standard form.

..... [1]

- 9 Solve.

$$\frac{1-p}{3} = 4$$

$p =$ [2]

- 10 Factorise completely.

$$2a + 4b - ax - 2bx$$

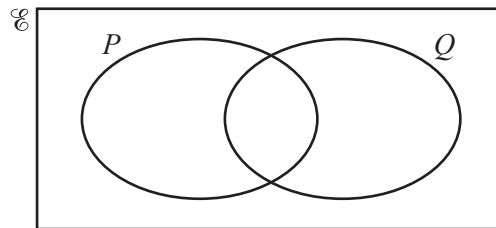
..... [2]

11 $A = (2\pi + y)x^2$

Rearrange the formula to make x the subject.

$x = \dots\dots\dots$ [2]

12



$n(E) = 20$, $n(P) = 10$, $n(Q) = 13$ and $n(P \cup Q)' = 5$.

Work out $n(P \cap Q)$.

You may use the Venn diagram to help you.

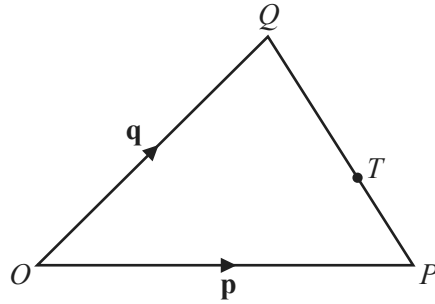
$n(P \cap Q) = \dots\dots\dots$ [2]

13 Simplify.

$$\frac{3+x}{9-x^2}$$

$\dots\dots\dots$ [2]

14

NOT TO
SCALE

O is the origin, $\overrightarrow{OP} = \mathbf{p}$ and $\overrightarrow{OQ} = \mathbf{q}$.
 $QT : TP = 2 : 1$

Find the position vector of T .
 Give your answer in terms of $\mathbf{p}$ and $\mathbf{q}$, in its simplest form.

..... [2]

15 Without using a calculator, work out $\frac{2}{3} \div 1\frac{1}{5}$.

You must show all your working and give your answer as a fraction in its simplest form.

..... [3]

16 (a) The length of the side of a square is 12 cm, correct to the nearest centimetre.

Calculate the upper bound for the perimeter of the square.

..... cm [2]

(b) Jo measures the length of a rope and records her measurement correct to the nearest ten centimetres.
 The upper bound for her measurement is 12.35 m.

Write down the measurement she records.

..... m [1]

- 17 (a) Find the value of $\left(\frac{1}{81}\right)^{-\frac{3}{4}}$.

..... [1]

- (b) Simplify. $\sqrt[3]{27t^{27}}$

..... [2]

- 18 Expand the brackets and simplify.
 $(2p+3)(3p-2)$

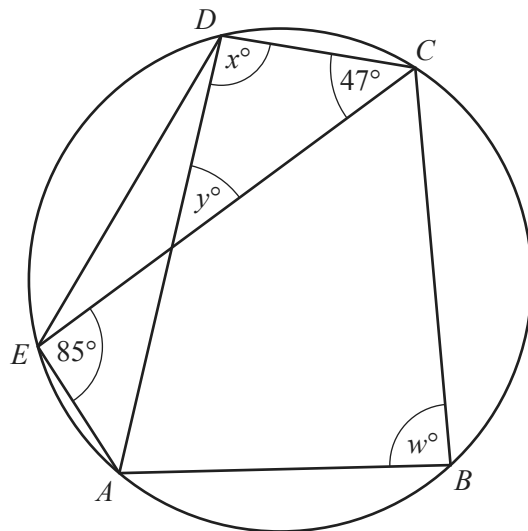
..... [3]

- 19 y is directly proportional to $(x-1)^2$.
 When $x = 3$, $y = 24$.

Find y when $x = 6$.

$y =$ [3]

20

NOT TO
SCALE

The points A , B , C , D and E lie on the circumference of the circle.
Angle $DCE = 47^\circ$ and angle $CEA = 85^\circ$.

Find the values of w , x and y .

$w =$

$x =$

$y =$ [3]

21 Write as a single fraction in its simplest form.

$$\frac{1}{y-1} - \frac{1}{y}$$

..... [3]

22 Find an expression for the n th term of each sequence.

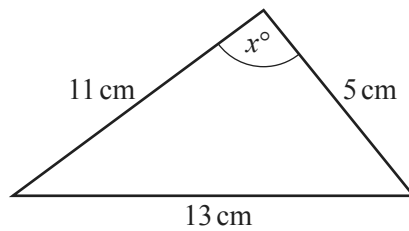
(a) 11, 7, 3, -1, ...

..... [2]

(b) 3, 6, 12, 24, ...

..... [2]

23

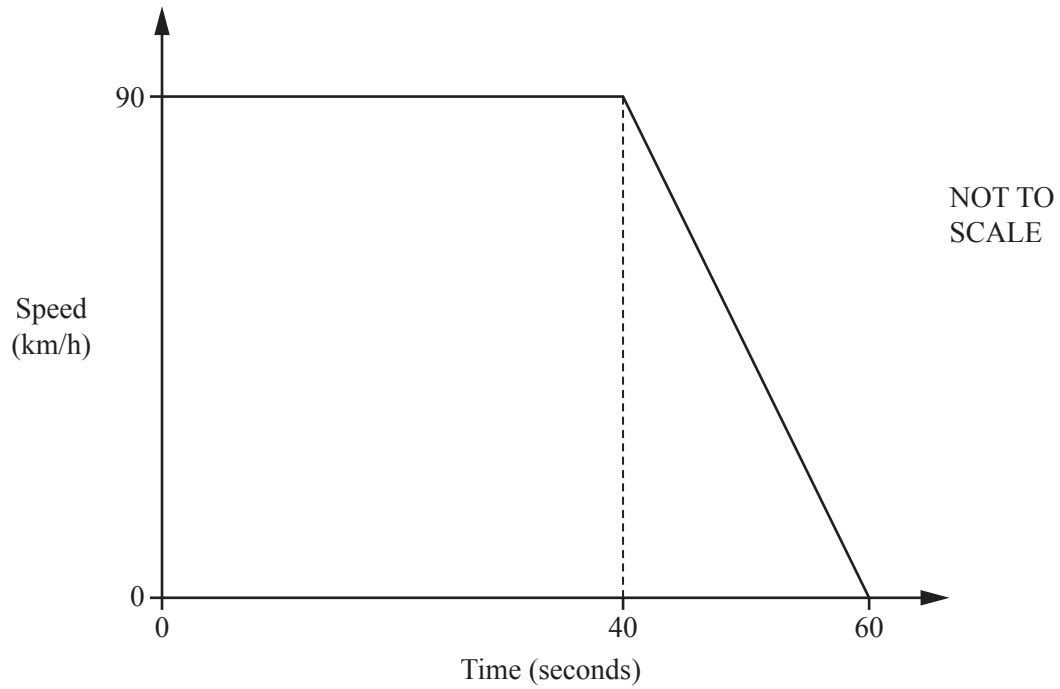


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Calculate the value of x .

$x =$ [4]

24



The diagram shows the speed–time graph for 60 seconds of a car journey.

- (a) Change 90 km/h to m/s.

..... m/s [2]

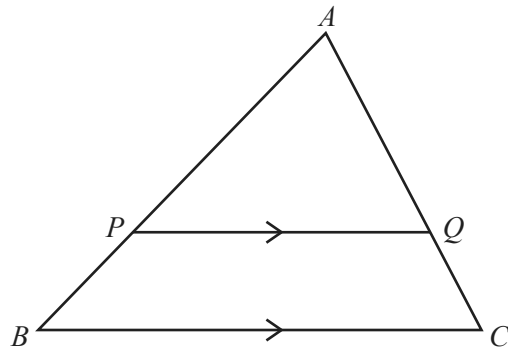
- (b) Find the deceleration of the car in m/s^2 .

..... m/s^2 [1]

- (c) Find the distance travelled, in metres, in the 60 seconds.

..... m [2]

25 (a)

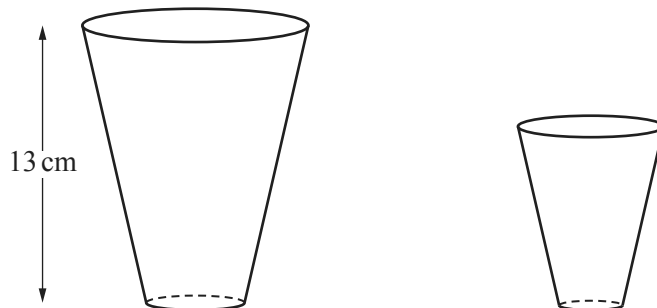
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In the diagram, PQ is parallel to BC .
 APB and AQC are straight lines.
 $PQ = 8$ cm, $BC = 10$ cm and $AB = 9$ cm.

Calculate PB .

$PB = \dots\dots\dots$ cm [2]

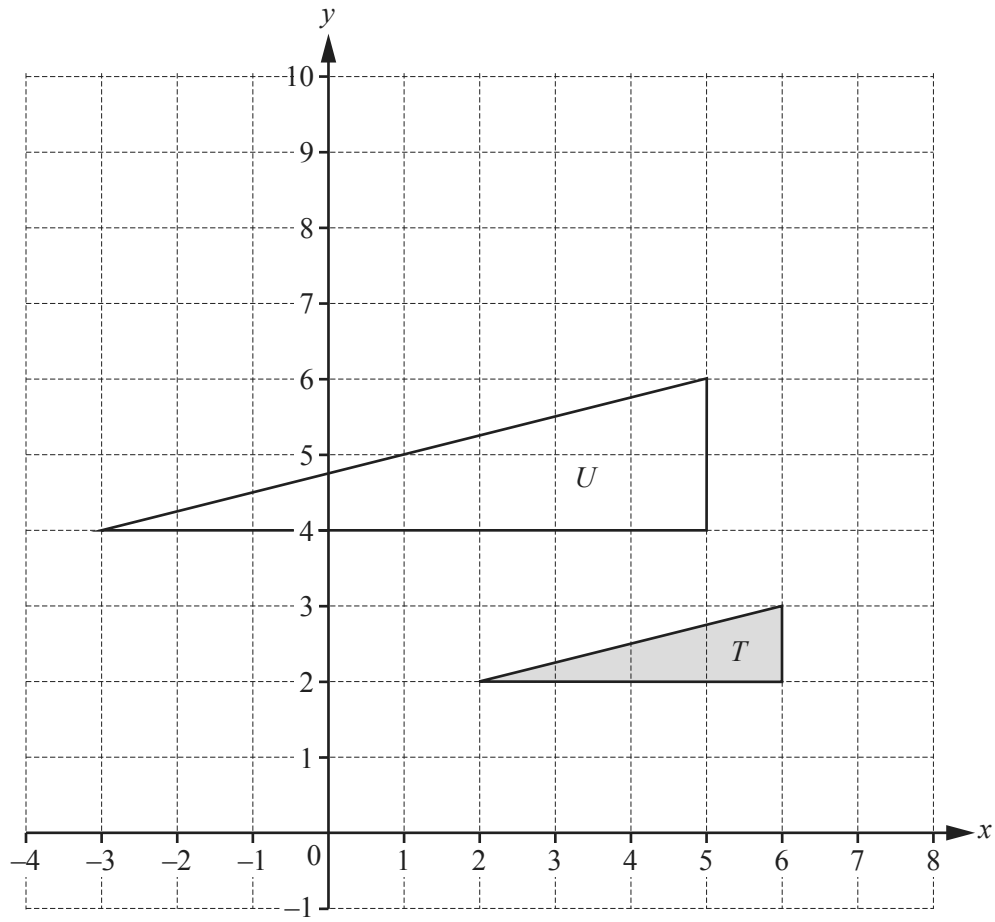
(b)

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The diagram shows two glasses which are mathematically similar.
 The larger glass has a capacity of 0.5 litres and the smaller glass has a capacity of 0.25 litres.
 The height of the larger glass is 13 cm.

Calculate the height of the smaller glass.

$\dots\dots\dots$ cm [3]



- (a) Describe fully the **single** transformation that maps triangle T onto triangle U .

.....
 [3]

- (b) On the grid, draw the image of triangle T after a rotation through 90° clockwise about the point $(7, 3)$.
 [3]

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